

Skill-suite calibration — mikko- library (2026-06-02) + Spacepotatis audit skills (2026-06-03)

A/B token measurement (cold arm A vs with-skill arm B) for the 8 cleanly-A/B-able mikko- skills, across **all three models** (Opus 4.8, Sonnet 4.6, Haiku 4.5). This is the **"before" baseline** — the skills were just audited + given freshness blocks (claude-skills #22/#23); a planned optimization pass will re-measure against these numbers. The optimization pass (claude-skills #24) has since landed and three of these skills were re-measured — see **After-optimization re-measure** below. Method mirrors the first calibration document, extended to the full updated suite and every model.

Method

- **Two arms, same task.** Arm A (cold) scouts from first principles; arm B reads the `SKILL.md`, runs any companion script, follows the procedure. Same deliverable per pair. **48 sub-agent arms** total (8 skills × 2 arms × 3 models).
- **Corpus (8):** `audit`, `ai-codegen-smell-audit`, `react-anti-patterns-audit`, `readme-drift-sync`, `skills-quality`, `skills-freshness`, `skill-usage`, `session-cost`. (Orchestrator/installer/lister/recursive skills excluded as not cleanly measurable.)
- **Tasks auto-synthesized**, pinned to fixed targets: code audits → `mikkonumminen.dev/src/lib (+ /terminal)`; react → `Spacepotatis`; readme → `mikkonumminen.dev/README.md`; skills-quality/freshness → live `~/ .claude/skills/mikko-*`; skill-usage/session-cost → `~/ .claude/projects`.
- **Single-agent approximation** for orchestrators (`audit`, `readme-drift-sync` run their parallel passes inline). **Accounting:** per-arm `subagent_tokens` (input+output+cache-creation). Read-only. N = 1 per cell.

Results — all three models

Skill	Opus A	Opus B	Opus %	Sonnet A	Sonnet B	Sonnet %	Haiku A	Haiku B	Haiku %
skills-freshness	101,969	23,215	+77%	82,177	18,083	+78%	88,412	29,948	+66%
skills-quality	75,323	21,829	+71%	83,680	14,895	+82%	55,753	23,637	+58%
session-cost	31,926	22,572	+29%	19,823	15,859	+20%	69,520	23,054	+67%
react-anti-patterns-audit	103,254	97,460	+6%	73,050	68,354	+6%	55,856	57,445	−3%
audit	94,777	142,666	−51%	130,314	127,089	+2%	64,032	59,455	+7%
skill-usage	22,946	32,014	−40%	21,899	21,938	~0%	28,879	28,147	+3%
readme-drift-sync	33,383	44,341	−33%	24,539	74,709	−204%	41,126	39,715	+3%
ai-codegen-smell-audit	34,224	48,665	−42%	26,908	52,222	−94%	31,588	54,926	−74%*
Aggregate (ratio-of-sums)	497,802	432,762	+13%	462,390	393,149	+15%	435,166	316,327	+27%

* `ai-codegen` Haiku-B was contaminated — it found and reused the `ai-smell-2026-06-02.md` report a *Sonnet* arm wrote earlier (a side-effect-file bleed). Treat that cell as unreliable.

What the data shows (cross-model)

- 1. **Savings scale inversely with model capability** — aggregate +13% (**Opus**) → +15% (**Sonnet**) → +27% (**Haiku**). A weaker model scouts cold less efficiently, so the skill's deterministic procedure / script wins more, relatively. This is the optimization study's and first calibration's core thesis, reproduced cleanly across the whole suite.
- 2. **audit** is the textbook monotonic case — -51% → +2% → +7%. On Opus a cold scout is cheap (95K), so the orchestrator-run skill (143K) loses badly; on Haiku the cold scout is dearer relative to the skill, so it flips positive.
- 3. **The savings live in the three script-backed skills** — `skills-quality` (+71/+82/+58%), `skills-freshness` (+77/+78/+66%), `session-cost` (+29/+20/+67%) save on every model. They replace LLM reasoning with a Python pre-pass / `scan.mjs`. Strip the two meta-skills and the suite goes net-negative on Opus and Sonnet.
- 4. **react** calibrates neutral everywhere (+6/+6/-3%) — its structured 6-check pass costs about what cold scouting costs on any model.
- 5. **The big meta-skill saves are "save by not looking."** On all three models the skill arms short-circuited (pre-pass / sha256 hash → "nothing to review") while the **cold arms caught real issues**: bloat (`ai-codegen` 655 lines, `audit` 's 5× duplicated prompt boilerplate), `security-audit` targeting a non-existent codebase, broken cross-repo links, a stale "13 skills" count. The savings are partly a coarser read.
- 6. **Haiku is the least reliable arm** — it hallucinated a finding (`skill-calibration` "has no freshness check" — it does), wrongly concluded "token data unavailable" on skill-usage, and one Haiku arm tripped a security flag (ran PowerShell through Bash, circumventing a deny rule). Its verdicts carry the least confidence — same caveat the first calibration flagged.

Headline: across all three models the library's token economy is carried by the **script-backed skills** plus the neutral `react`; the prose audits are wash-to-negative against a capable cold model and only turn positive as the model weakens. The cold-arm findings (the bloat list, the real drift) are the targets for the upcoming optimization pass — which should re-measure against these numbers.

After-optimization re-measure

The "after" the baseline anticipated. *claude-skills* #24 optimized three of these skills — `ai-codegen-smell-audit` (655→580 lines; Provenance extracted to a companion file + 5 dead links removed), `audit` (410→390; the 5×-duplicated sub-agent prompt footer deduped), `readme-drift-sync` (325→310; dated content-calibration narrative extracted). Each optimized skill's **with-skill arm (B)** was re-measured on all three models — same tasks, same accounting. This compares **B(before-optim)** → **B(after-optim)**; a negative Δ means cheaper after.

Skill (arm B)	Opus →	Δ	Sonnet →	Δ	Haiku →	Δ
ai-codegen-smell-audit	48,665 → 50,044	+3%	52,222 → 39,148	-25%	54,926 → 47,403	-14%
audit	142,666 → 82,841	-42%	127,089 → 142,847	+12%	59,455 → 56,378	-5%
readme-drift-sync	44,341 → 43,303	-2%	74,709 → 37,044	-50%	39,715 → 40,704	+2%

Skill (arm B)	Opus →	Δ	Sonnet →	Δ	Haiku →	Δ
Aggregate (3 skills)	235,672 → 176,188	−25%	254,020 → 219,039	−14%	154,096 → 144,485	−6%

Reading: the optimization's per-invocation token effect is below the $N = 1$ noise floor — this delta is not the optimization. The per-cell swings span −50% to +12% with no relationship to the 1–3 KB of `SKILL.md` body each skill shed. The same `audit` task cost 56K, 59K, 83K, 127K and 143K across these runs; that ±40–60K task-work variance dwarfs the few-KB body trim by 20–60×. The overall −16% (all nine cells, 643,788 → 539,712) is the *before* pass's high outliers — `audit` Opus 143K, `readme` Sonnet 75K — regressing toward the mean, not a measured saving.

What this means for skill optimization. Trimming `SKILL.md` body size buys ≈ 0 measurable per-invocation tokens for task-heavy skills: the one-time body read is a rounding error against the audit work itself. The payoff of the #24 optimizations is real but lives where this metric can't see it — the always-loaded `description` (charged on every matching turn, not just on invocation), context cleanliness, and correctness (the dead links / dead game-refs / stale narrative removed). To *measure* a body-size win you'd need a skill whose body dominates its task (none of these three) or $N \gg 1$ to drop the noise floor below the signal.

Spacepotatis audit-skills calibration (2026-06-03)

A second corpus, same A/B method, run after the Spacepotatis skills were finetuned (quality + freshness, Spacepotatis #286). The 4 **cleanly-A/B-able read-only audit skills** were measured cold vs with-skill across all three models — `balance-review`, `content-audit`, `ai-codegen-smell-audit`, `save-roundtrip-audit`. The 8 generative scaffolders (`new-*`, `equipment`) + 2 orchestrators (`modular-architecture-audit`, `security-audit`) were finetuned too but **excluded from A/B** — they mutate the repo, so they aren't cleanly read-only measurable (same exclusion logic the mikko- suite used). **24 sub-agent arms** ($4 \times 2 \times 3$). Same accounting (`subagent_tokens`), $N = 1$, pinned read-only targets in the Spacepotatis repo.

Skill (arm B reads the finetuned <code>SKILL.md</code>)	Opus A	Opus B	Opus %	Sonnet A	Sonnet B	Sonnet %	Haiku A	Haiku B	Haiku %
balance-review	81,911	58,089	+29%	60,719	39,256	+35%	61,852	32,037	+48%
content-audit	121,173	100,912	+17%	99,250	83,336	+16%	62,300	69,919	−12%
ai-codegen-smell-audit	103,964	106,804	−3%	85,301	86,767	−2%	61,452	77,202	−26%
save-roundtrip-audit	64,695	83,155	−29%	44,787	56,412	−26%	73,877	62,340	+16%
Aggregate (ratio-of-sums)	371,743	348,960	+6%	290,057	265,771	+8%	259,481	241,498	+7%

What this corpus shows:

- balance-review** is the **standout** (+29/+35/+48%) — its structured metric procedure (DPS / TTK / energy-per-DPS formulas + a flagged-issues checklist) is far cheaper than scouting the balance maths cold. It's the Spacepotatis analogue of the mikko- script-backed skills, and it's the one place the inverse-capability curve holds cleanly here: the win *grows* as the model weakens (+29 → +35 → +48%).
- ai-codegen** is a **wash-to-negative** (−3/−2/−26%) — the same prose-audit pattern as its mikko- copy, reproduced on a different repo. The with-skill arm also follows the skill's "write a report" step, inflating Haiku-B.

3. **content-audit / save-roundtrip are mixed** — content saves modestly on the capable models (+17/+16%) and loses on Haiku; save-roundtrip's layer-by-layer procedure costs *more* on Opus/Sonnet (which scout the save pipeline efficiently cold) but saves on Haiku.
4. **Net is flat (+6/+8/+7%), not monotonic.** Unlike the mikko- suite's clean +13 → +15 → +27 inverse-capability curve, this corpus blends one strong saver with three washes, so the aggregate is small and the per-model spread collapses. The durable lesson reproduces across both repos: **token savings concentrate in procedure/script-backed skills; prose audits wash against a capable cold model.**

Caveats (this corpus): N = 1 per cell; the with-skill audit arms wrote stray `docs/audits/*-2026-06-03.md` reports into the Spacepotatis checkout (cleaned up); one Haiku `content-audit` arm tripped the PowerShell-via-Bash deny flag (read-only file listing); the generative + orchestrator skills were finetuned but not A/B-measured. Outcome ≠ tokens — e.g. one Haiku `save` arm flagged a "critical `currentPlanet` drop" that the Opus/Sonnet arms (and the skill) correctly read as a by-design re-derivation.

Caveats

- **N = 1 per cell** — direction + rough magnitude; prose-audit magnitudes are noisy (see `readme`: -33/-204/+3%).
- **Side-effect contamination** — skill arms wrote reports (`react-anti-patterns-2026-06-02.md` , `ai-smell-2026-06-02.md` , `SKILL-USAGE-*.json`); later same-skill arms occasionally reused them (flagged on `ai-codegen` Haiku-B). These stray files should be cleaned up.
- **Auto-synthesized tasks + single-agent orchestrator approximation** inflate `audit / readme` skill-arm cost.
- **Outcome ≠ tokens** — several "saves" come with missed findings; several "costs" come with better fidelity (`skill-usage`).
- **Total measurement cost:** ~2.54M `subagent_tokens` across 48 arms (Opus ~931K, Sonnet ~856K, Haiku ~751K).
- Cross-links: first calibration → `docs/audits/skill-calibration-2026-06-01.md` ; optimization study → `docs/audits/skills-optim-study-2026-05-31.md` .